

Delay Event-Triggered Control for Stability Analysis of Complex Networks

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Abstract—This brief is concerned with exponential stability of delayed complex networks. Different from existing literature, an extended control is proposed, namely delay event-triggered control, in which the update of control signals is determined by the event-triggered mechanism. Both continuous event-triggered mechanism and periodic event-triggered mechanism are developed by continuous measuring and periodic sampling, respectively. The continuous event-triggered mechanism with waiting time is proposed to ensure that the existence of the minimum inter-execution time has a positive minimum value. The upper bound of the sampling period is given for the periodic event-triggered mechanism and an estimation about sampling/execution error is obtained to assess the network states. Subsequently, by employing the Lyapunov method and graph-theoretic approach, stability criteria for delayed complex networks under delay event-triggered control are established. Ultimately, the analytical results are applied into single-link robot arm systems and a corresponding numerical example is given to illustrate the effectiveness and feasibility of theoretical claims.

Index Terms—Delay event-triggered control, delayed complex networks, periodic event-triggered control.

I. INTRODUCTION

IN THE past decades, complex networks have attracted a tremendous amount of attention on account of their wide applicability in many fields, including economics, biology, physics, and many other areas (see [1], [2] and their references therein). Since time delays are always unavoidable in many practical complex networks and may cause oscillatory behaviors or network instability, it is necessary to investigate the influence of time delay of complex networks. As a typical dynamic properties, the study on the stability of delayed complex networks has received much attention (see, e.g., [3], [4], [5]) in recent years.

Sampled-data control of large-scale complex networks has been always a hot topic due to its numerous merits. In terms of triggering mechanisms, sampled-data control can be classified into two categories. The first one is time-triggered control, which releases sampling signals at certain predefined moments (see, e.g., [6], [7]). It can be seen that time-triggered control is essentially time-driven and usually requires a fixed sampling period. This feature may cause unnecessary data information to be transmitted through the network, causing huge pressure

on the actuators. In contrast, the other manner, event-triggered control, only releases the sampling signals when the predefined event-triggered conditions are violated, and the control signal is updated (see, e.g., [8], [9], [10]). Therefore, compared with time-triggered control, event-triggered control can significantly reduce the number of triggers to reduce the waste of computing resources.

However, due to the time delays of signal transmission between sensors, controllers and executors in practice, information acquisition and the updated control inputing to the system are not completed at the same time (e.g., it takes 1.28 seconds for a radio signal from the moon to reach the earth [11]). In other words, time delays are not only exist in systems, its also should be considered in control strategy. Recently, many scholars have made some progress on the delay time-triggered control of complex networks [11], [12]. In [11], [12], the stabilisation of hybrid systems via delay feedback control and intermittent delay feedback control based on discrete-time observations was considered, where the sampling instants are pre-specified. In contrast to time-triggered control, event-triggered control has been introduced in which the control signal is only activated after pre-defined event occurs, rather than at the fixed sampling time instants [13]. Avoiding Zeno behavior is a fundamental but challenging problem in delay event-triggered control. And many remarkable results on this problem have been reported in the literature [8], [9], [10], [14], [15], [16], [17], [18]. For systems without time delays, a hybrid event-triggered control strategy, which consists the state-dependent term and time-dependent term, was proposed in [8]. The feature of this strategy is that the interevent intervals may approach zero as time goes. In [9] the method introduced to exclude Zeno behavior, is to analyze the dynamics of $\frac{\epsilon(t)}{|x(t)|}$, in which $x(t)$ is the real-time state of system and $\epsilon(t)$ is the measurement error. However, the above method for avoiding Zeno behavior cannot be directly extended to delayed networks, primarily because the state can be zero in a finite time due to the existence of time delays [16]. In [17], a delay-dependent threshold for a class of nonlinear time-delay systems was presented. The proposed strategy in [18] excludes Zeno behavior, however, stability of the control systems may not be maintained. Therefore, how to extend the existing results of the time-triggered control systems to the delay event-triggered control systems and design the event-triggered mechanism to ensure the Zeno-freeness is extremely worth considering. Inspired by the above discussions, we aim to explore the DE-TC for delayed complex networks. The dominant contributions of this brief are listed as follows:

- Compared with delay time-triggered control [11], [12], in which the sampling time interval is fixed or determined in advance, the delay event-triggered control which is generated by pre-designed event-triggered conditions is only activated after the event occurs. Therefore we take the delay event-triggered control into consideration.

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- In contrast to [17], [18], the continuous event-triggered mechanism with waiting time is proposed, in which the waiting time can be chosen by a simple calculation. Meanwhile, the preset waiting time ensures that there exists a positive minimum value for the minimum inter-execution time, which means Zeno behavior can be strictly excluded.
- The periodic event-triggered mechanism is first developed for delay event-triggered control systems. In addition, the upper bound of the sampling period is given and sampling/execution error is estimated and governed to evaluate the performance of the networks.

Notations: Let $\mathbb{R}$ stand for the set of real number and $\mathbb{R}^+ = [0, \infty)$, $\mathbb{R}^m$ represents the m -dimensional Euclidean space and $\|\cdot\|$ is the Euclidean norm of a matrix or a vector in $\mathbb{R}^m$. Set $\mathcal{M} = \{1, 2, \dots, M\}$. For function $x(t)$, define the right and upper Dini derivation as $D^+x(t) = \lim_{h \rightarrow 0^+} \frac{x(t+h) - x(t)}{h}$. Besides, $C([- \tilde{\tau}, 0]; \mathbb{R}^m)$ represents the space of continuous functions $\tilde{\psi} : [- \tilde{\tau}, 0] \rightarrow \mathbb{R}^m$ with norm $\|\tilde{\psi}\| = \sup_{-\tilde{\tau} \leq s \leq 0} |\tilde{\psi}(s)|$. A weighted digraph containing M vertices is described as $\mathcal{G} = (V, \mathcal{E}, (d_{ab})_{M \times M})$, where $V = \{v_1, v_2, \dots, v_M\}$ and $\mathcal{E}$ denotes the node set and the edge set respectively, $D = (d_{ab})_{M \times M}$ is the weighted matrix of $\mathcal{G}$. For simplicity, the digraph can be rewritten as $(\mathcal{G}, D)$. Moreover, $\mathcal{G}$ is said to be strongly connected if for any two vertices of $\mathcal{G}$, there exists a path consisting of directed edges from one to the other.

II. MODEL DESCRIPTION

In this section, we consider the following delayed complex networks with time-varying coupling

$$\dot{q}_a(t) = \tilde{f}_a(q_a(t), q_a(t - \tau_1), t) + \sum_{b=1}^M d_{ab}(t) \times H_{ab}(q_a(t - \tau_2), q_b(t - \tau_3)) + u_a(t), \quad a \in \mathcal{M}, \quad (1)$$

where $q_a \in \mathbb{R}^m$, $q(t) = (q_1^T, q_2^T, \dots, q_M^T)^T \in \mathbb{R}^{Mm}$ is the state of network (1), function $f_a : \mathbb{R}^m \times \mathbb{R}^m \times \mathbb{R}^+ \rightarrow \mathbb{R}^m$ is continuous, H_{ab} stands the coupling function. And $d_{ab}(t)$ represents the time-varying coupling strength, which is bounded, that is $0 \leq d_{ab}(t) < d_{ab}$. If the b th node has an influence on the a th node, then $d_{ab}(t) > 0$, otherwise $d_{ab}(t) = 0$. Moreover, τ_1 is the internal delay, τ_2 is the self-delay and τ_3 denotes the neighboring delay.

Next, the delay event-triggered control is designed as follows

$$u_a(t) = -k_a q_a(t_m - \tau_4), \quad t \in [t_m, t_{m+1}), \quad (2)$$

in which $k_a > 0$ represents the control gain of delay event-triggered control, τ_4 is the delay in the delay event-triggered control system. And we give a structure chart in Fig. 1.

Remark 1: As a combination of periodic sampling mechanism and event-triggered control, the proposed control strategy is based on the requirement of intermittent monitored, which greatly reduces the update of control signals and the consumption of communication resources. In contrast to the time-triggered control based on a fixed sampling interval, the existence of the event trigger threshold ensures that the control update is based on the network state, which further improves the flexibility of the control strategy

Remark 2: As the inevitable factors of real networks, time delays are considered in many studies [15], [19]. Among various types of time delays, those occur inside the nodes are called internal delay, and both self-delays and neighboring delays that exist between nodes are collectively called coupling

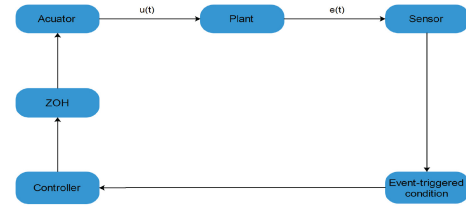


Fig. 1. Structure chart of delayed complex networks based on delay event-triggered control.

delay (or external delay). However, the above work either ignore certain types of delays or consider some delays as equal. For the sake of generality, the networks in this brief are considered with different internal delays, self-delays and neighboring delays.

Assumption 1: There exist $\beta_{a1}, \beta_{a2}, \epsilon_{ab} > 0$ such that

$$\|\tilde{f}_a(q_a, \tilde{q}_a, t) - \tilde{f}_a(z_a, \tilde{z}_a, t)\| \leq \beta_{a1} \|q_a - z_a\| + \beta_{a2} \|\tilde{q}_a - \tilde{z}_a\|,$$

and

$$\|H_{ab}(q_a, q_b, t) - H_{ab}(\tilde{q}_a, \tilde{q}_b, t)\| \leq \epsilon_{ab} (\|q_a - \tilde{q}_a\| + \|q_b - \tilde{q}_b\|), \quad (3)$$

where $q_a, \tilde{q}_a, z_a, \tilde{z}_a \in \mathbb{R}^m$.

Assumption 2: The digraph (G, Γ) is strongly connected, where $\Gamma = (\Gamma_{ab})_{m \times m}$, $\Gamma_{ab} = \max\{d_{ab}^2 \epsilon_{ab}^2, d_{ab} \epsilon_{ab}\}$. Let c_a is the cofactor of the a th diagonal element of the Laplacian matrix of digraph (G, Γ) satisfying $\underline{c} \leq c_a \leq \bar{c}$, where $\underline{c}$ and $\bar{c}$ are positive constants.

III. MAIN RESULTS

In this section, sufficient criteria will be given to guarantee the exponential stability of delayed complex networks (1) based on delay event-triggered control (2). Two types of event-triggered mechanisms, namely: 1) periodic event-triggered mechanism and 2) continuous event-triggered mechanism, are designed.

A. Periodic Event-Triggered Mechanism

Firstly, the execution time instants are determined as

$$t_{m+1} = \inf \left\{ l\Delta \geq t_m \mid \sum_{a=1}^M c_a \|q_a(l\Delta - \tau_4) - q_a(t_m - \tau_4)\|^2 \geq \delta \sum_{a=1}^M c_a \|q_a(l\Delta - \tau_4)\|^2 \right\}, \quad (4)$$

where $\delta > 0$ is a triggering parameter and $\Delta > \tau_4$ is the sampling period. Define $\tau^* = \max\{\tau_1, \tau_2, \tau_3, \Delta + \tau_4\}$. To simplify the proof, we define $x(t) = (x_1^T, x_2^T, \dots, x_M^T)^T$, $x_a(t) = q_a(\eta_t - \tau_4) - q_a(t)$ and $\tilde{x}_a^m = q_a(\xi_t - \tau_4) - q_a(\eta_t - \tau_4)$, where $\eta_t = \max\{l\Delta \leq t, l \in \mathbb{N}\}$, $\xi_t = \max\{t_m \leq t, m \in \mathbb{N}^+\}$.

Lemma 1: Assume that Assumptions 1-2 hold. When $\|\tilde{x}^m\|^2 \neq 0$, then for any $t \in [l\Delta, (l+1)\Delta)$, the following estimation can be obtained

$$\sum_{a=1}^M c_a \|q_a(t) - q_a(\xi_t - \tau_4)\|^2 \leq Z_1(\Delta) \sum_{a=1}^M c_a \|q_a(t)\|^2 + Z_2(\Delta) \sup_{t-2\tau^* \leq s \leq t} \sum_{a=1}^M c_a \|q_a(s)\|, \quad (5)$$

where $\varphi_1 = \max_{a \in \mathcal{M}} \{6(\Delta + \tau_4)(k^2(\delta + 1) + 2\beta_{a1}^2)\}$, $\varphi_2 = \max_{a \in \mathcal{M}} \{12(\Delta + \tau_4)\beta_{a1}^2\}$, $\varphi_3 = \max_{a \in \mathcal{M}} \{6(\Delta + \tau_4)\beta_{a2}^2\}$, $\varphi_4 = \max_{a \in \mathcal{M}} \{6(\Delta + \tau_4)M \sum_{b=1}^M \Gamma_{ab}\}$, $\varphi_5 = (\varphi_1 + \varphi_3 + 2\varphi_4)\tau^*$,

$Z_1(\Delta) = \frac{2\delta}{\lambda_1}$, $Z_2(\Delta) = 2(1 + \frac{\delta}{1-\lambda_1}) \varphi_5 \exp\{\varphi_2(\Delta + \tau_4)\}$, $k = \max_{a \in M} \{k_a\}$, in which $0 < \lambda_1 \leq 1$.

Proof: Firstly, make that $\tilde{f}_a(q_a, \tilde{q}_a, t) = \tilde{f}_a(q_a, \tilde{q}_a, 0)$, $H_{ab}(\tilde{q}_a, \tilde{q}_b, t) = H_{ab}(\tilde{q}_a, \tilde{q}_b, 0)$, for any $q_a, \tilde{q}_a, \tilde{q}_b \in \mathbb{R}^m$, $t \in [-2\tau^*, 0)$. Besides, for any initial conditions, $q \in C([- \tau^*, 0]; \mathbb{R}^{Mm})$, let $q(t) = q(-\tau^*)$, for $t \in [-2\tau^*, \tau^*)$.

When $\|\tilde{x}^m\|^2 \neq 0$, we can obtain that

$$\begin{aligned} & \sum_{a=1}^M c_a \|q_a(t) - q_a(\xi_t - \tau_4)\|^2 \\ & \leq 2 \sum_{a=1}^M c_a \|x_a(t)\|^2 + 2 \sum_{a=1}^M c_a \|\tilde{x}_a^m\|^2 \\ & \leq 2 \sum_{a=1}^M c_a \|x_a(t)\|^2 + 2\delta \sum_{a=1}^M c_a \|q_a(\eta_t - \tau_4)\|^2 \\ & \leq 2 \left(1 + \frac{\delta}{1-\lambda_1}\right) \sum_{a=1}^M c_a \|x_a(t)\|^2 + \frac{2\delta}{\lambda_1} \sum_{a=1}^M c_a \|q_a(t)\|^2, \end{aligned}$$

where $0 < \lambda_1 < 1$. Then, based on Assumption 1, one has

$$\begin{aligned} \|x_a(t)\|^2 & \leq \left\| \int_{l\Delta-\tau_4}^t \left(\tilde{f}_a(q_a(s), q_a(s-\tau_1), s) + \sum_{b=1}^M d_{ab}(s) \right. \right. \\ & \quad \times \left. \left. H_{ab}(q_a(s-\tau_2), q_b(s-\tau_3), t) + k_a q_a(\xi_s - \tau_4) \right) ds \right\|^2 \\ & \leq (\Delta + \tau_4) \int_{l\Delta-\tau_4}^t (3 \|\tilde{f}_a(q_a(s), q_a(s-\tau_1), s)\|^2 \\ & \quad + 3M \sum_{b=1}^M (d_{ab}(s))^2 \|H_{ab}(q_a(s-\tau_2), q_b(s-\tau_3), s)\|^2 \\ & \quad + 3 \|k_a q_a(\xi_s - \tau_4)\|^2) ds \end{aligned} \quad (6)$$

According to delay event-triggered control (4), it can be derived that

$$\begin{aligned} & \sum_{a=1}^M c_a \|q_a(\xi_t - \tau_4)\|^2 \\ & \leq 2 \sum_{a=1}^M c_a \|\tilde{x}_a^m\|^2 + 2 \sum_{a=1}^M c_a \|q_a(\eta_t - \tau_4)\|^2 \\ & \leq 2(\delta + 1) \sum_{a=1}^M c_a \|q_a(\eta_t - \tau_4)\|^2. \end{aligned} \quad (7)$$

Then, combined with (6) and (7), it can yield that

$$\begin{aligned} \sum_{a=1}^M c_a \|x_a(t)\|^2 & \leq \int_{l\Delta-\tau_4}^t \sum_{a=1}^M c_a 6(\Delta + \tau_4) (2\beta_{a1}^2 \|x_a(s)\|^2 + \beta_{a2}^2 \\ & \quad \times \|q_a(s-\tau_1)\|^2 + M \sum_{b=1}^M \Gamma_{ab} (\|q_a(s-\tau_2)\|^2 \\ & \quad + \|q_a(s-\tau_3)\|^2) + (k_a^2(\delta + 1) \frac{\bar{c}}{c} + 2\beta_{a1}^2) \\ & \quad \times \|q_a(\eta_s - \tau_4)\|^2) ds + 6M(\Delta + \tau_4) \int_{l\Delta-\tau_4}^t \sum_{a=1}^M c_a \\ & \quad \times \sum_{b=1}^M \Gamma_{ab} (\|q_b(s-\tau_3)\|^2 - \|q_a(s-\tau_3)\|^2) ds. \end{aligned}$$

Then, it follows that

$$\sum_{a=1}^M c_a \|x_a(t)\|^2 \leq \varphi_5 \exp\{\varphi_2(\Delta + \tau_4)\} \sup_{t-2\tau^* \leq s \leq t} \sum_{a=1}^M c_a \|q_a(s)\|^2. \quad (8)$$

Therefore, the inequality (5) can be easily obtained. The proof is completed. ■

Theorem 1: Suppose that Assumptions 1-2 hold. The exponential stability of system (1) can be achieved under delay event-triggered control when condition $\Lambda_5 > \sum_{i=1}^3 \Lambda_i + \frac{k^2}{2\lambda_2} Z_2(\Delta)$ is satisfied, in which $\Lambda_0 = \min_{a \in M} \{-2\beta_{a1} - \beta_{a2} - 2\lambda_2 - \sum_{b=1}^M 2\Gamma_{ab} + 2k_a\}$, $\Lambda_1 = \max_{a \in M} \{\beta_{a2}\}$, $\Lambda_2 = \Lambda_3 = \max_{a \in M} \{\sum_{b=1}^M \Gamma_{ab}\}$, $\Lambda_5 = \Lambda_0 - \frac{k^2}{2\lambda_2} Z_1(\Delta)$.

Proof: First, we define

$$V(q, t) = \sum_{a=1}^M c_a V_a(q_a, t),$$

where $V_a(q_a, t) = \|q_a\|^2$. Then, we can calculate that

$$\begin{aligned} D^+ V(q(t), t) & = \sum_{a=1}^M c_a \left[2q_a^T(t) \left(\tilde{f}_a(q_a(t), q_a(t-\tau_1), t) \right. \right. \\ & \quad \left. \left. + \sum_{b=1}^M d_{ab} H_{ab}(q_a(t-\tau_2), q_b(t-\tau_3), t) + u_a(t) \right) \right] \end{aligned} \quad (9)$$

Case I: When $\|\tilde{x}_a^m\|^2 = 0$, according to the Assumption 1, we obtain

$$\begin{aligned} & 2q_a^T(t) \tilde{f}_a(q_a(t), q_a(t-\tau_1), t) \\ & \leq 2\beta_{a1} \|q_a(t)\|^2 + 2\beta_{a2} \|q_a(t)\| \|q_a(t-\tau_1)\| \\ & \leq (2\beta_{a1} + \beta_{a2}) \|q_a(t)\|^2 + \beta_{a2} \|q_a(t-\tau_1)\|^2 \end{aligned} \quad (10)$$

By combining (9) and (10), it can yield that

$$\begin{aligned} D^+ V(q(t), t) & \leq \sum_{a=1}^M c_a [(2\beta_{a1} + \beta_{a2} + 2\lambda_2 + 2 \sum_{b=1}^M \Gamma_{ab} - 2k_a) \\ & \quad \times \|q_a(t)\|^2 + \beta_{a2} \|q_a(t-\tau_1)\|^2 + \frac{k_a^2}{2\lambda_2} \|q_a(t) \\ & \quad - q_a(\eta_t - \tau_4)\|^2] + \sum_{b=1}^M \Gamma_{ab} (\|q_a(t-\tau_2)\|^2 + \|q_a(t-\tau_3)\|^2) \\ & \quad + \sum_{a,b=1}^M c_a \Gamma_{ab} (\|q_b(t-\tau_3)\|^2 - \|q_a(t-\tau_3)\|^2) \end{aligned} \quad (11)$$

Based on [20, Th. 2.2], we can obtain

$$\sum_{a=1}^M \sum_{b=1}^M c_a \Gamma_{ab} (\|q_b(t-\tau_3)\|^2 - \|q_a(t-\tau_3)\|^2) = 0. \quad (12)$$

By substituting (12) to (11), we have

$$\begin{aligned} D^+ V(q(t), t) & \leq -\Lambda_0 \sum_{a=1}^M c_a \|q_a(t)\|^2 + \sum_{i=1}^3 \sum_{a=1}^M \\ & \quad \times \Lambda_i c_a \|q_a(t-\tau_i)\|^2 + \frac{k^2}{2\lambda_2} \tilde{Z}_2(\Delta) \sup_{t-2\tau^* \leq s \leq t} V(q(s), s) \end{aligned}$$

Case II: When $\|\tilde{x}_a^m\|^2 \neq 0$, similarly it can be obtained that

$$\begin{aligned} D^+ V(q(t), t) & \leq -\Lambda_5 \sum_{a=1}^M c_a \|q_a(t)\|^2 + \sum_{i=1}^3 \sum_{a=1}^M \Lambda_i c_a \|q_a(t-\tau_i)\|^2 \\ & \quad + \frac{k_a^2}{2\lambda_2} Z_2(\Delta) \sup_{t-2\tau^* \leq s \leq t} V(q(s), s) \end{aligned}$$

Thus, based on [21, Lemma 3], it can yield that

$$V(q(t), t) \leq \sup_{-2\tau^* \leq s \leq 0} V(q(s), s) \exp\{-\mu t\}, \quad t \geq 0.$$

in which μ is the smallest real root of formula $\mu = \Lambda_5 - \sum_{i=1}^3 \Lambda_i \exp\{2\tau_i \mu\} - \frac{k^2}{2\lambda_2} Z_2(\Delta) \exp\{2\tau^* \mu\}$. Then, according to $\underline{c} \|q(t)\|^2 \leq \sum_{a=1}^M c_a V_a(q_a(t), t) \leq \bar{c} \|q(t)\|^2$, we have

$$\begin{aligned} \|q(t)\|^2 &\leq \frac{\bar{c}}{\underline{c}} \sup_{-2\tau^* \leq s \leq 0} \|q(s)\|^2 \exp\{-\mu t\} \\ &\leq \frac{\bar{c}}{\underline{c}} \sup_{-2\tau^* \leq s \leq 0} \|q(s)\|^2 \exp\{-\mu t\}, \quad t \geq 0. \end{aligned}$$

Therefore, the exponential stability of system (1) can be achieved. The proof is completed. ■

Remark 3: In Theorem 1, the Tree-Cycle Identity [20] is employed to handle the coupling terms between subsystems. This method has been considerably applied in the research of delayed complex networks under intermittent control or impulsive control (see, e.g., [4], [6]). However, different from the previous researches [4], [6], we study the delay event-triggered control for the first time.

Remark 4: When considering the use of event-triggered control to study the dynamic properties of the system, how to avoid the Zeno behavior becomes a key issue. In [9], [22], some authors excluded the Zeno behavior by obtaining the minimum lower bound of the inter-execution time through calculating that the quotient of the measurement error and real-time system state. In [10], the authors designed a distributed dynamic event-triggered control in which a dynamic term is introduced to exclude the Zeno behavior. However, for delayed systems, the existing methods are not applicable. The proposed strategy in [18] rules out Zeno behavior, however, while boundedness and attractivity of the systems under the proposed event-triggered control algorithm can be guaranteed, stability of the control systems may not be maintained. In this brief, inspired by [23], [24], [25], [26], the positive minimum inter-execution time is obtained by introducing periodic sampling mechanism into event-triggered control which can avoid the appearance of the Zeno phenomenon naturally.

B. Continuous Event-Triggered Mechanism

To avoid Zeno behavior, we design the continuous event-triggered mechanism with waiting time

$$\begin{aligned} t_{m+1} = \inf \left\{ t \geq t_m + \Delta \left| \sum_{a=1}^M c_a \|q_a(t) - q_a(t_m - \tau_4)\|^2 \right. \right. \\ \left. \left. \geq \delta_2 \sum_{a=1}^M c_a \|q_a(t)\|^2 \right\}, \end{aligned}$$

where $\Delta > \tau_4$ acts as a waiting time to exclude the Zeno behavior and $\delta_2 > 0$ is a triggering parameter.

Theorem 2: Suppose that Assumptions 1-2 hold. System (1) under continuous event-triggered mechanism with waiting time can achieve exponential stability, if $\Lambda_6 > \sum_{i=1}^3 \Lambda_i + \max\{\frac{k^2}{2\lambda_2} \tilde{Z}_2(\Delta), \frac{\delta_2 k^2}{\lambda_2}\}$, where $\Lambda_6 = \min_{a \in M} \{-2\beta_{a1} - \beta_{a2} - 2\lambda_2 - \sum_{b=1}^M 2d_{ab} \epsilon_{ab} + 2k_a\}$.

Proof: Similar to the proof of Theorem 1, we also need to discuss the two cases. When $t \in [t_m, t_m + \Delta)$, it follows from Case I of Theorem 1 that (11) can be obtained. When $t \in [t_m + \Delta, t_{m+1})$, based on continuous event-triggered mechanism with waiting time, we have $\sum_{a=1}^M c_a \|q_a(t) - q_a(t_m - \tau_4)\|^2 \leq \delta_2 \sum_{a=1}^M c_a \|q_a(t)\|^2$. Thus, it can be obtained that

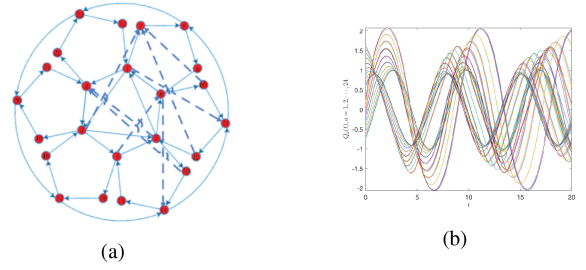


Fig. 2. (a) A graph $\mathcal{G}$ with 24 vertices. (b) Trajectories of system (14) (without control).

$\|q(t)\|^2 \leq \frac{\bar{c}}{\underline{c}} \sup_{-2\tau^* \leq s \leq 0} \|q(s)\|^2 \exp\{-\mu_2 t\}$, $t \geq 0$. The proof is completed. ■

Remark 5: For continuous event-triggered mechanism with waiting time, the event-triggered mechanism adopts a two-stage approach: once the execution occurs, the trigger will remain paused for a fixed period of time; only after the pause, an event-triggering condition is continuously monitored to determine the triggering instants. It is unnecessary to monitor the network state and check the event-triggered mechanism in $(t_m, t_m + \Delta)$. Thus, comparing with continuous event-triggered mechanisms, event-triggered mechanism with waiting time can reduce the communication and computation resources.

IV. NUMERICAL EXAMPLES

Consider a single-link robot arm system as follows

$$\tilde{J} \ddot{q}_a(t) + B \dot{q}_a(t) + mgL \sin(q_a(t)) = 0, \quad a \in M, \quad (13)$$

where $q_a(t)$, $\dot{q}_a(t)$ and $\ddot{q}_a(t)$ represent the angle, angular velocity and angular acceleration, respectively. $\tilde{J}$ represents the total rotational inertia of the link and the motor, B is damping coefficient. Besides, m and L are total mass and distance between the joint axis and the link center of the mass, g is acceleration of gravity.

Then, consider that coupling factor, and time delays into system (13), the following system can be obtained

$$\begin{aligned} \dot{Q}_a(t) = F_a(Q_a(t), Q_a(t - \tau_1), t) + \sum_{b=1}^M d_{ab}(t) H_{ab} \\ \times (Q_a(t - \tau_2), Q_b(t - \tau_3), t) + u_a(t), \quad a \in \mathcal{M}, \quad (14) \end{aligned}$$

where the relevant event-triggered mechanisms are similarly defined in Section III, $Q_a(t) = (q_a(t), \tilde{q}_a(t))^T$ and $F_a(Q_a(t), Q_a(t - \tau_1), t) = (\tilde{q}_a(t), -\frac{1}{\tilde{J}}(B\tilde{q}_a(t) + mgL(\sin(q_a(t)) + \sin(q_a(t - \tau_1))))^T$, $H_{ab}(Q_a(t - \tau_2), Q_b(t - \tau_3), t) = (0, h_{ab}(q_a(t - \tau_2), q_b(t - \tau_3), t))^T$.

Take $m = 0.2$, $B = 0.1$, $g = 9.8$, $L = 2.5$, $\tilde{J} = 11.1$. Then, we can estimate that $\|F_a(P_a(t), P_a(t - \tau_1), t) - F_a(Q_a(t), Q_a(t - \tau_1), t)\| \leq \lambda_a \|P_a(t) - Q_a(t)\| + \theta_a \|P_a(t - \tau_1) - Q_a(t - \tau_1)\|$. Based on these data, it can be derived that $\lambda_a = 1.0003$, $\theta_a = 0.4414$. Take $M = 24$ and we model system (14) on digraph $(\mathcal{G}, (\Gamma_{ab})_{24 \times 24})$, which can be observed in Fig. 2 (a).

From the framework, we can figure out that digraph $(\mathcal{G}, (\Gamma_{ab})_{M \times M})$ is strongly connected and the non-zero elements of matrix $(d_{ab}(t))_{24 \times 24}$ are provided as follows: $d_{1,6} = d_{2,1} = d_{2,21} = d_{3,1} = d_{7,1} = d_{8,11} = d_{10,6} = d_{17,16} = 0.015 + 0.001 \exp\{-t\}$, $d_{3,2} = d_{5,12} = d_{8,7} = d_{13,5} = d_{14,17} = d_{16,4} = d_{18,17} = d_{21,20} = d_{22,2} = d_{23,8} = d_{23,22} = d_{24,23} = 0.024 + 0.002 \exp\{-t\}$, $d_{3,18} = d_{4,3} = d_{8,23} = d_{9,8} = d_{14,13} = d_{19,3} = d_{20,23} = 0.023 + 0.0015 \exp\{-t\}$, $d_{1,24} = d_{5,4} = d_{6,9} = d_{11,14} = d_{15,14} = d_{20,19} =$

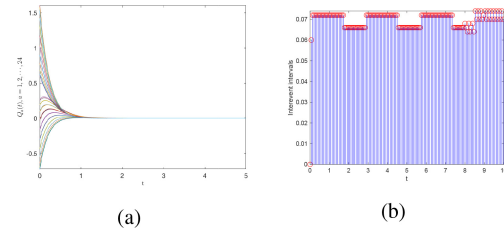


Fig. 3. (a) The event-triggered instants under periodic event-triggered mechanism. (b) The trajectories of system (14) under periodic event-triggered mechanism.

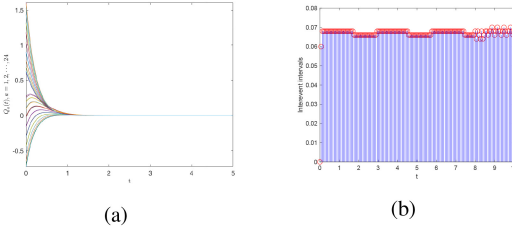


Fig. 4. (a) The event-triggered instants. (b) The trajectories of system (14) under continuous event-triggered mechanism.

$0.013 + 0.0014 \exp\{-t\}$, $d_{1,5} = d_{4,15} = d_{6,5} = d_{11,10} = d_{17,20} = 0.014 + 0.0013 \exp\{-t\}$,

Then, select functions $h_{ab} = 0.01 \sin(q_a(t - \tau_2)) + 0.01 \sin(q_b(t - \tau_3))$, and then we can choose $\epsilon_{ab} = 0.1$ which implies Assumption 1 holds. Meanwhile, we take $\tau_1 = \tau_2 = 0.008$, $\tau_3 = 0.006$. Clearly, when there is no control input, i.e., $u(t) = 0$, system (14) is unstable, which is depicted in Fig. 2 (b).

In order to make system (14) stable, we consider the delay event-triggered control described in (2). Take $\tau_4 = 0.002$, $\Delta = 0.006$, $\delta_1 = \delta_2 = 0.2$ and $k_a = 3.5$ for $a = 1, 2, \dots, 24$. Simple calculation yields $\Lambda_5 > \sum_{i=1}^3 \Lambda_i + \frac{k^2}{2\lambda_2} Z_2(\Delta)$, which implies that all the conditions in Theorem 2 are satisfied. The trajectories of system (14) under periodic event-triggered mechanism and continuous event-triggered mechanism are given in Figs. 3–4 (b), which indicates that the proposed delay E-TC can stabilize the system (14). Besides, to better illustrate the process of triggering, the event-triggered instants generated by periodic event-triggered mechanism and continuous event-triggered mechanism are presented in Fig. 3–4 (a), which implies that the Zeno behavior can be excluded in our results.

V. CONCLUSION

We studied the delay event-triggered control problem for delayed complex networks. Both continuous event-triggered mechanism and periodic event-triggered mechanism were developed by continuous measuring and periodic sampling, respectively. By applying the Lyapunov method, graph theory and inequality techniques, the sufficient conditions are provided. In the future, we will consider intermittent delay event-triggered control to achieve the stability of Markovian switching complex networks.

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